Material Selection and Design for a Tennis Racket

Cristina Cuesta Bret

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Introduction

With technological advances and the change in the way tennis is played, new materials are being considered to design a better tennis racket. Tennis has changed as a sport in the past two decades. Players are more athletic, hit the ball with more force and thus the speed of the ball increases and rallies are longer, so rackets need to be light. Tennis racket manufacturers must keep up with all these changes so that the equipment doesn't fail and doesn't cause injuries due to force and torque.

What is interesting about this product is that the pro players use different, usually customized, rackets that have very different design specifications to the ones sold to the public. So why do pro players need their rackets modified, why do amateurs and beginners use different ones, why are there so many models? Most importantly, what material is better and why.

Application

A tennis racket is used to hit a ball over a net towards the opponent. There is a specific movement technique used to hit the ball that directly affects ball speed, torque and overall effectiveness of the shot and the racket is a big part to hitting effectively. Tennis players usually have a hard time choosing a racket because there are so many models, each with different specifications that can affect how the player hits the ball and the likelihood of getting injured. If the racket is too heavy or too long, the player can easily get injured. It is important that rackets produce the required ball speed with the least amount of effort while keeping elbow, shoulder and wrist safety.

Tennis rackets used to be made from wood, today they are made entirely from graphite or use graphite as a composite blend with carbon fiber, Kevlar, titanium, fiberglass, or tungsten (1).

Background and Objectives

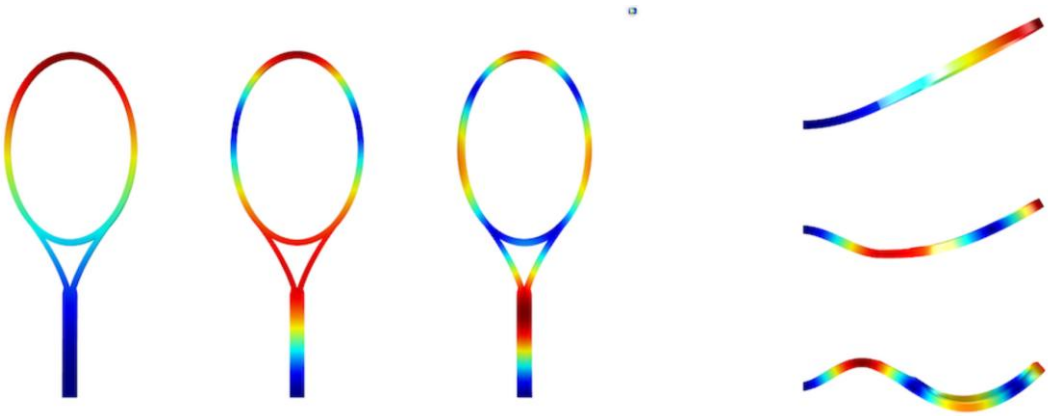

Figure 1: COMSOL simulation of forces experienced by the racket frame (4)

Tennis rackets experience many types of forces such as moments, torque, torsion, tension, and impact forces. It is also important to consider that the player's arm also experiences such forces and choosing the correct material and racket dimensions can decrease these forces. As Figure 1 shows, there are many ways a racket can vibrate due to all the forces it experiences. The most common way a tennis racket can fail is when the strings have the wrong tension, and the ball hits the wrong part of the frame. This will cause cracks on the frame due to vibrations and shock. The racket should be able to hold the strings to a specific tension, it should also be able to withstand repeated impact with the ball.

Proper rackets must be within length range and head frame size, specified by the ITF (International Tennis Federation). A racket must not exceed 29 in (73.66 cm) in length, including the handle; it must not exceed 12 ½ in in width (1). Today, tennis rackets are

made in different weights and sizes that match the player using them. Usually, an adult amateur player will play with an 11 oz racket.

Formed using molds, resin layups and heat, the racket should be lightweight and strong/resistant. This is why graphite and carbon fiber are most used because they are light and strong materials. It also must be durable. (5)

Different tests are done before a racket is sold. These tests include tensile strength, dampening shock, and elongation to replicate the ball hitting the strings.

Material Selection

I will be comparing young's modulus, density, tensile strength and hardness for carbon fiber, tungsten, titanium, and wood. Tennis rackets are very complex and require a strong material but not too hard so that it doesn't damage the player when used. Rackets used today are made of a combination of two or more of the materials I am comparing, but since I can't mix them in Ansys, I will be evaluating them separately and assuming that the racket will only be made of one material.

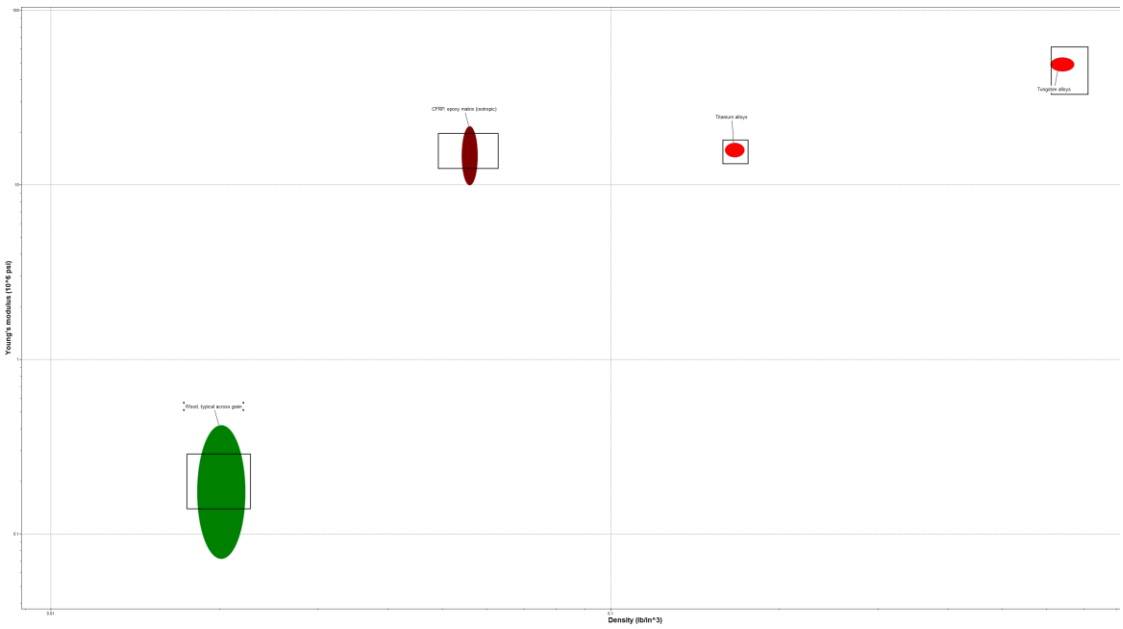

Figure 2: Young's Modulus vs Density of possible materials

In Figure 2, I am comparing young's modulus with density of the material. Wood seems to be the least dense but is also the one with the smallest Young's modulus which means that the material is more ductile than the other materials. Tungsten is the hardest material and the densest, while carbon fiber and titanium have similar Young's modulus, titanium is much denser. From this graph alone, we can consider carbon fiber or titanium to be the best options because they're lighter than wood and are still strong. Wood would still be an okay option if we only look at this graph.

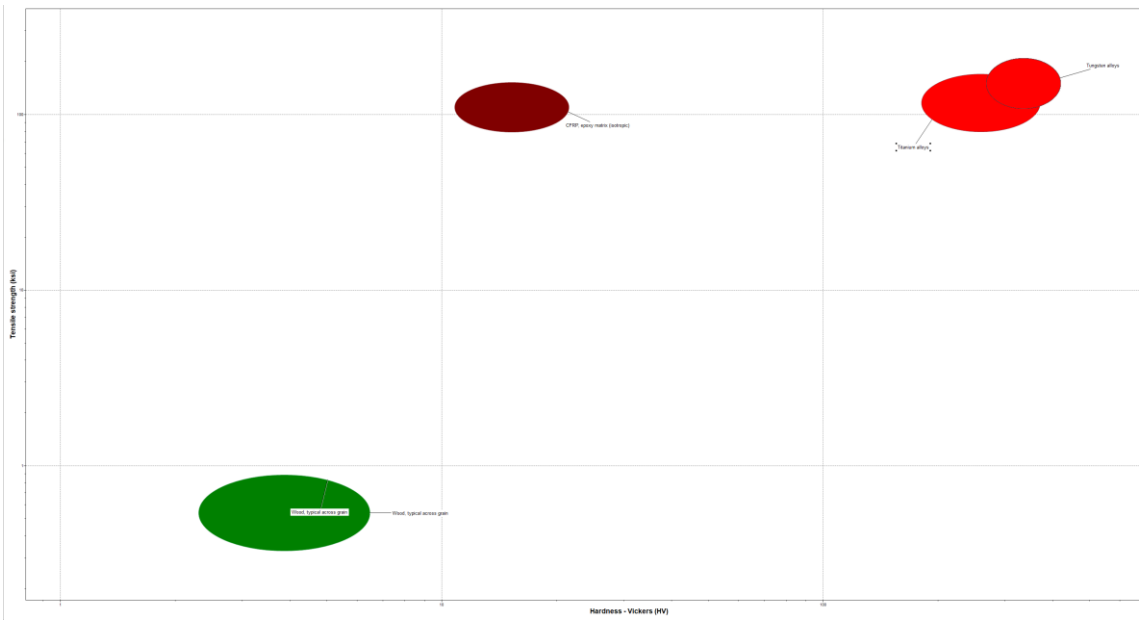

Figure 3: Tensile Strength vs Hardness

In Figure 3, I plotted Tensile Strength vs Hardness. The ideal material that we are looking for is a strong material with a high tensile strength but also that is not too hard so that it doesn't injure the player. We can clearly see that wood is significantly less hard than the other materials and its tensile strength is much smaller. Wood is not hard so it could snap more easily than the other materials, which means that it could break if the loads applied are too large. Once again, carbon fiber seems to be the best option as it has a high tensile strength and is not too hard. Tungsten and titanium are very similar and they are both very hard metals.

Table 1: Material Testing summary

Material	Young's Modulus (10^6 psi)	Density (lb/in^3)	Tenslie Strength (ksi)	Hardness (HV)
Carbon Fiber	10-21.8	0.0542-0.0578	79.8-152	10.8-21.5
Wood	1.26-2.18	0.017-0.0226	8.93-18.6	2.97-8.28
Titanium	14.5-17.4	0.16-0.173	80.1-170	181-370
Tungsten	45-53.7	0.611-0.672	109-209	268-419

The best choice according to these tests is carbon fiber, it is the lightest, yet strongest material out of the four that I selected. Wood is not strong enough. Tungsten and titanium are too hard and too heavy for them to be the only material used to make a tennis racket.

Eco-Audit with Granta

The results after performing the eco-audit on Granta show that making the carbon fiber itself would be where the most energy is consumed and when the CO2 footprint is the largest. This makes sense since the product chosen doesn't require any other type of fossil fuel or electric energy after it reaches the player. For the study, I have taken the Head headquarters in Austria as the factory the racket was made at and to deliver it in Spain, which is where I bought mine. I have also used compression molding as the manufacturing process since that is what is used by Head.

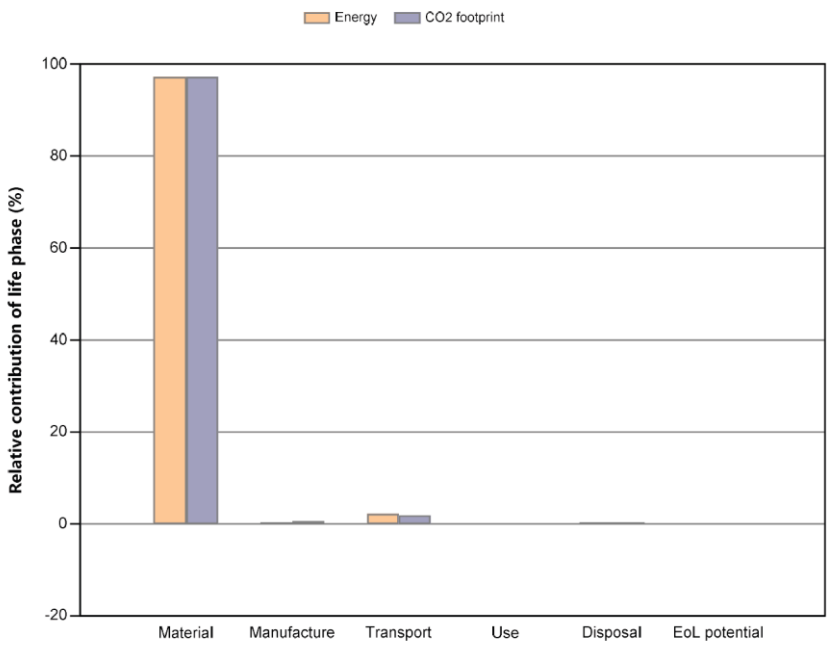

Figure 4: Energy and CO2 footprint of the racket

Table 2: Summary of Energy and CO2 footprint

Phase	Energy (J)	Energy (%)	CO2 footprint (kg)	CO2 footprint (%)
Material	2.18e+08	97.4	15.3	97.4
Manufacture	1.05e+06	0.5	0.0839	0.5
Transport	4.68e+06	2.1	0.312	2.0
Use	0	0.0	0	0.0
Disposal	6e+04	0.0	0.0042	0.0
Total (for first life)	**2.24e+08**	**100**	**15.7**	**100**
End of life potential	0		0	

Table 3: Product Specifications

Material:

Summary

Component	Material	Recycled content* (%)	Part mass (kg)	Qty.	Total mass (kg)	Energy (J)	%
Racket Frame	CFRP, epoxy matrix (isotropic)	Virgin (0%)	0.3	1	0.3	2.2e+08	100.0
Total				**1**	**0.3**	**2.2e+08**	**100**

*Typical: Includes 'recycle fraction in current supply'

***User-defined material

References

1) Miller, N. (2022, July 5). *What are tennis rackets made of?*. Discovery UK. https://www.discoveryuk.com/how-its-made/what-are-tennis-rackets-made-of/
2) *Racquet science - using physics to help select a racquet*. ATP Equipment Rules. (n.d.). https://www.racquetresearch.info/atp
3) *Racquet science - using physics to help select a racquet*. Introduction to Racquet Science. (n.d.). https://www.racquetresearch.info/sevencri
4) *The physics of tennis racket sweet spots*. COMSOL. (n.d.). https://www.comsol.com/blogs/the-physics-of-tennis-racket-sweet-spots/
5) YouTube. (2020, May 1). *How a head tennis racquet is made -- behind the scenes in Kennelbach, Austria!*. YouTube. https://www.youtube.com/watch?v=n62fqfBUuIY